Mathematical Methods

- The Essential mathematical tools to understand the physics are **Vector analysis** and **Elementary calculus**

Scalars and Vectors

- **Scalars**: Physical quantities completely described by their magnitude (a number and a unit). Examples: Mass, time, temperature. **They** follow ordinary algebra.
- · **Vectors**: Physical quantities that require both magnitude and direction for a complete description. Examples: Displacement, velocity, force. **They** follow vector algebra. Representation: A vector is represented by a directed line segment or as $\bar{A}$, with magnitude $|\bar{A}|$.

Mathematical Methods (conts.)

Types:

1. Zero Vector (Null vector):

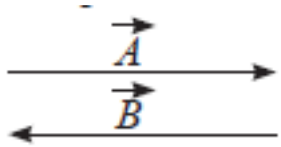
A vector having zero magnitude with a particular direction (arbitrary). Symbolically it is represented as $\bar{O}$. (e.g Velocity vector of a stationary particle is a zero vector, The acceleration vector of an object moving with uniform velocity is a zero vector).

2. resultant vector:

The resultant of two or more vectors is that single vector, which produces the same effect, as produced by all the vectors together.

3. negative vector (Opposite Vectors):

A negative vector of a given vector is a vector of the same magnitude but opposite in direction to that of the given vector.

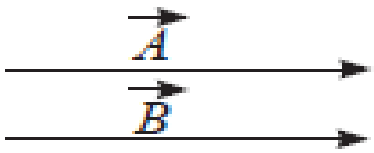


$\bar{A}$ Is Negative Vector to $\bar{B}$

Mathematical Methods (Conts.)

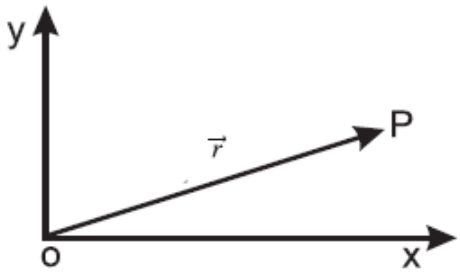
4. equal vectors:

Two vectors A and B representing same physical quantity are said to be equal if and only if they have the same magnitude and direction.



5. position vector.

A vector which gives the position of a particle at a point with respect to the origin of a chosen coordinate system is called the position vector of the particle.



$\bar{r}$ = $\overline{OP}$ is the position vector of the particle present at P.

6. Unit Vector:

A vector having unit magnitude in a given direction is called a unit vector in that direction.

- A vector with a magnitude of 1, used to specify direction. It is denoted as $\hat{u}$ (e.g., $\hat{i}$, $\hat{j}$, $\hat{k}$ for x, y, z axes). $\hat{u} = \bar{A} / |\bar{A}|$.

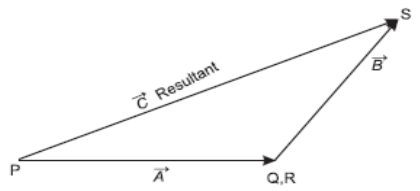
Mathematical Methods (conts.)

Vector Operations:

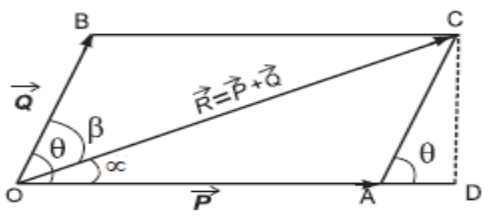
- Multiplication by a Scalar: Multiplying a vector by a scalar changes its magnitude but not its direction (unless the scalar is negative, which reverses the direction).

A. Addition and Subtraction:

1.**Triangle Law**: If two vectors are represented by two sides of a triangle taken in order, their resultant is given by the third side taken in the opposite order.

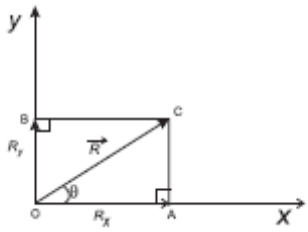


2. Parallelogram Law: If two vectors acting at a point are represented by two adjacent sides of a parallelogram, the resultant is given by the diagonal passing through that point.



3**. Vector addition:** is commutative ($\bar{A} + \bar{B} = \bar{B} + \bar{A}$) and associative ($\bar{A} + (\bar{B} + \bar{C}) = (\bar{A} + \bar{B}) + \bar{C}$).

4. **Resolution of Vectors**: Splitting a vector into components along predetermined axes (e.g., x, y, z). For a vector $\bar{R}$ in the x-y plane:



$R_x = R \cos\theta$

$R_y = R \sin\theta$

$R = \sqrt{(R_x^2 + R_y^2)}$

$\theta = \tan^{-1}(R_y / R_x)$

Mathematical Methods (Conts.)

B. Multiplication of Vectors

1. Scalar (Dot) Product: Result is a scalar.

$\bar{A} \cdot \bar{B} = |A||B| \cos\theta$ (where θ is the angle between them).

- It gives the product of the magnitude of one vector and the component of the other in its direction.
- Application: Work ($W = \bar{F} \cdot \bar{s}$).

2. Vector (Cross) Product: Result is a vector.

$\bar{A} \times \bar{B} = |A||B| \sin\theta\ \hat{u}$ (where $\hat{u}$ is a unit vector perpendicular to the plane containing $\bar{A}$ and $\bar{B}$, given by the right-hand thumb rule).

- Its magnitude is equal to the area o f the parallelogram formed by the two vectors.
- Application: Torque ($\bar{\tau} = \bar{r} \times \bar{F}$), Magnetic force on a moving charge.
- **Unit Vectors**

A unit vector is a dimensionless vector with a magnitude of exactly 1

Unit vectors are used to specify a direction and have no other physical significance

The symbols $\hat{\imath}, \hat{\jmath}, and\ \hat{k},$ represent unit vectors.

They form a set of mutually perpendicular vectors in a right-handed coordinate system

The magnitude of each unit vector is 1:

$$|\hat{\imath}| = |\hat{\jmath}| = |\hat{k}| = 1$$

A_x is the same as $A_x\hat{l}\ and\ A_x$is the same as $A_y\hat{\jmath}$ etc.

The complete vector can be expressed as:

$$\vec{A} = A_x\hat{l} + A_y\hat{\jmath}$$

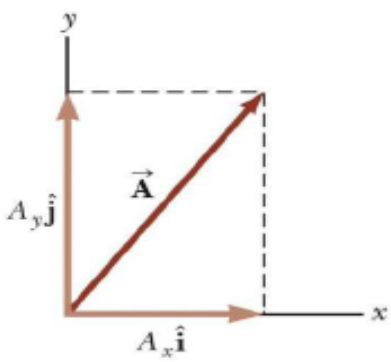
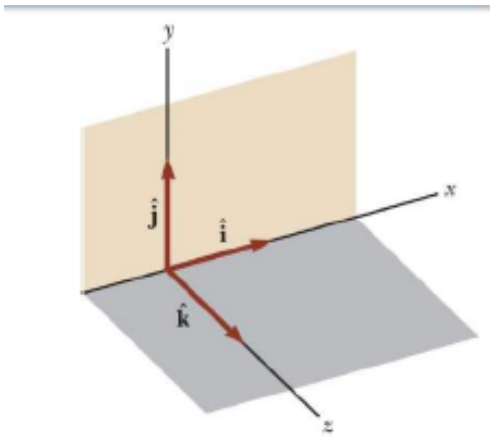


Diagram: Shows unit vectors $\hat{\imath}, \hat{\jmath}, and\ \hat{k}$ in 3D coordinate system



Two-Dimensional Extension

Using: $\vec{R} = \vec{A} + \vec{B}$

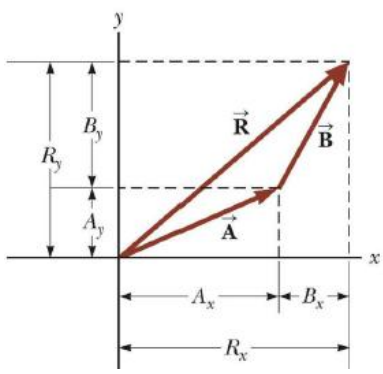
Then : $\vec{R} = \left(A_x\hat{l} + A_y\hat{\jmath}\right) + \left(B_x\hat{l} + B_y\hat{\jmath}\right)$

$$\vec{R} = (A_x + B_x)\hat{l} + \left(A_y + B_y\right)\hat{\jmath}$$

$$\vec{R} = \left(R_x\hat{l} + R_y\hat{\jmath}\right)$$

So : $R_x = A_x + B_x\ and R_y = A_y + B_y$

$R = \sqrt{R_x^2 + R_y^2}$ $\qquad$ $\theta = tan^{-1}\frac{R_x}{R_y}$



Three-Dimensional Extension:

$$\vec{R} = \left(A_x\hat{i} + A_y\hat{j} + A_z\hat{k}\right) + \left(B_x\hat{i} + B_y\hat{j} + B_z\hat{k}\right)$$

$$\vec{R} = (A_x + B_x)\hat{i} + \left(A_y + B_y\right)\hat{j} + (A_z + B_z)\hat{k}$$

$$\vec{R} = \left(R_x\hat{i} + R_y\hat{j} + R_z\hat{k}\right)$$

$$R_x = A_x + B_x + \text{ and } R_y = A_y + B_y \text{ and } R_z = A_z + B_z$$

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2}$$

$$\theta = \cos^{-1}\frac{\vec{A}.\vec{B}}{|\vec{A}|.|\vec{B}|} = \frac{A_xB_x + A_yB_y + A_zB_z}{\sqrt{(A_xB_x)^2 + +\left(A_yB_y\right)^2 + (A_zB_z)^2}}$$

Example

Two vectors are given by $A = 3i - 2j$ and $B = -i - 4j$. Calculate (a) $\vec{A} + \vec{B}$, (b) $\vec{A} - \vec{B}$, (c) $\left|\vec{A} + \vec{B}\right|$, (d) $\left|\vec{A} - \vec{B}\right|$, and (e) the direction of $\vec{A} + \vec{B}$ and $\left|\vec{A} - \vec{B}\right|$.

(a) $\vec{A} + \vec{B} = (3i - 2j) + (-i - 4j) = 2i - 6j$

(b) $\vec{A} - \vec{B} = (3i - 2j) - (-i - 4j) = 4i + 2j$

(c) $\left|\vec{A} + \vec{B}\right| = \sqrt{2^2 + (-6)^2} = 6.32$

(d) $\left|\vec{A} - \vec{B}\right| = \sqrt{4^2 + 2^2} = 4.47$

(e) For $\vec{A} + \vec{B}$, $\theta = \tan^{-1}(-6/2) = -71.6^\circ = 288^\circ$

For $\vec{A} - \vec{B}$, $\theta = \tan^{-1}(2/4) = 26.6^\circ$

- **Mathematical Methods (Conts.)**
- **Introduction to Calculus**

- Calculus is the study of continuous (not discrete) changes in mathematical quantities.
- The basics of the two branches of calculus namely **differential** and **integral** calculus

1. Differential Calculus: Deals with the rate of change.

- The derivative of a function $y = f(x)$ with respect to x is dy/dx. It represents the instantaneous rate of change of y with respect to x, or the slope of the tangent to the curve y vs. x.
- Application: Velocity ($v = dx/dt$) is the derivative of position with respect to time. Acceleration ($a = dv/dt$) is the derivative of velocity.

2. Integral Calculus: It is the reverse process of differentiation (anti-derivative).Deals with the accumulation of quantities.

- The integral of a function gives the area under its curve. The definite integral $\int f(x)dx$ from a to b gives the total accumulation between a and b.
- Application: Displacement is the integral of velocity with respect to time ($s = \int v\, dt$). Work done by a variable force is $W = \int F\, ds$.